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Microphase-dependent reinforcement–weakening crossover by nanoparticles in polyurethane

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ABSTRACT

In this study, the effect of nanoparticle-induced microphase modulation on the mechanical behavior of polyurethane (PU) was investigated. Phase-separated PU microstructures with embedded nanoparticles were implemented in a coarse-grained molecular dynamics environment, and their tensile properties were systematically evaluated. The results theoretically revealed that nanoparticles exert contrasting effects on the mechanical performance of PU, which is governed by its intrinsic microphase morphology. The enhancement factor decreases significantly (1.61 to 0.82 in elastic modulus and 1.30 to 0.66 in toughness) as the hard segment content increases from 13.3 wt.% to 43.5 wt.%. Nanoparticles with high surface polarity preferentially attract hard segments because of their favorable interfacial energetics. Furthermore, selective adsorption induces a compositional gradient, wherein the adjacent region beyond the interfacial layer exhibits a depletion of hard segments. Virtual peel tests also confirmed that the interfacial hard domains exhibit significantly enhanced cohesive strength and deformation resistance. Nanoparticles cause spatially confined, mechanically robust "hard segment islands" to form within the PU matrix. Such microstructural modulation yields divergent mechanical outcomes—reinforcement in globular phases but mechanical degradation in elongated or bicontinuous phases. These findings underscore the importance of morphology-aware design strategies for engineering PU nanocomposites.

1. Introduction

Polyurethane (PU) is a block copolymer in which the hard and soft segments are arranged alternately to simultaneously provide high strength and flexibility under mechanical deformation [1–8]. Because of the significant thermodynamic incompatibility between these segments, PU undergoes spontaneous microphase separation to form distinct nanostructured domains [9–20]. The degree of microphase separation and resultant morphology are critically dependent on the relative proportions of hard and soft segments [21–28]. As the hard segment content in the PU molecule increases, the microstructure transitions from isolated globular to discrete elongated to interconnected bicontinuous networks. These morphological transitions strongly influence stress distributions and load transfer pathways within PU [29], allowing for a wide range of mechanical properties, ranging from elastomeric and compliant to glassy and rigid. Such structural adaptability underscores the applicability of PU in diverse industrial sectors, including pressure-sensitive adhesives, automotive components, biomedical scaffolds, and structural coatings [30–35].

Incorporation of nanoparticles has emerged as a universal strategy to overcome the limitations of the intrinsic mechanical properties of polymer matrices. However, in the case of PUs, such reinforcement strategies face significant challenges because of the inherently heterogeneous microstructures of PUs. The defining mechanical feature of PU-based nanocomposites is the strong dependence of reinforcement efficiency on the microphase morphology. Experimental studies have demonstrated that the nanoparticle-induced enhancement is pronounced at low hard segment concentrations (30–35 wt. %), whereas the inclusion of nanoparticles at higher hard segment fractions (50–55 wt. %) produces marginal improvements or even a deterioration in tensile strength and toughness [36–38] (**Fig. S1**). This contrasting composition-dependent effect implies that the reinforcement capability of nanoparticles is intricately governed by the phase-separated morphology of the PU matrix.

To explore the origin of the above phenomenon, experimental techniques have been introduced to investigate the microphase morphology affected by the hard segment fraction and nanoparticles. Fourier transform infrared spectroscopy (FT-IR) analysis indicated

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extensive hydrogen bonding at the nanoparticle–matrix interface [39]. Polar surface groups on the nanoparticles associated preferentially with urethane linkages, producing localized enrichment of hard segments adjacent to the particle surface. Consistent with this interfacial adsorption, diffraction features suggested increased ordering of hard-rich domains near the interface. Complementary atomic force microscopy phase imaging showed that nanoparticle addition enhances the extent of microphase separation, manifested by higher phase contrast and sharper hard–soft domain boundaries [40–42]. These observations support a mechanism in which nanoparticles act as heterogeneous nucleation sites that promote thermodynamic demixing between hard and soft segments. Nevertheless, the mechanistic basis for nanoparticle-induced, morphology-dependent reinforcement remains incompletely resolved. This is because (1) conventional approaches have limited ability to track dynamic, in-situ microstructural responses under mechanical load, and (2) most measurement techniques are surface-sensitive rather than bulk-representative.

Microstructure modeling and virtual evaluation conducted in molecular dynamics (MD) simulation environments hold promise for overcoming above limitations. Molecular rearrangements driven by differences in energetic preferences allow for a detailed analysis of composition-dependent, time-resolved microstructural evolution [43–45]. Hydrogen-bond formation, interfacial reorganization, and strain-field distribution under mechanical loads are explicitly captured within a subcontinuum framework. Furthermore, bulk-representative cells with periodic boundaries isolate true volumetric responses from surface interference, thereby enabling estimation of intrinsic constitutive laws unconfounded by artifacts. Overall, the MD framework links interfacial energetics to domain morphology and macroscopic mechanical response, delivering predictive and morphology-aware guidance for PU nanocomposite design.

Motivated by the above, we explored the molecular origin of the microphase-dependent reinforcement efficiency of PU nanocomposites in silico. PUs and nanocomposites with various hard segment fractions were implemented in a coarse-grained MD environment. The spatial distribution of PU segments and their associated mechanical responses owing to nanoparticle incorporation were closely analyzed. We focused on two opposing effects that occur simultaneously when incorporating nanoparticles into a PU matrix, namely, localized mechanical reinforcement and network continuity disruption. These two influencing factors yield competing modulations in mechanical outcomes, resulting in markedly different reinforcement efficiencies that are dependent on the microphase morphology. The results theoretically explain the importance of design strategies for microphase to appropriately utilize the nanoparticle-induced reinforcement.

This work examines how nanoparticle insertion alters the mechanics of microphase-separated polyurethane by inducing interfacial hard-segment “islands” and a compliant depletion layer that redistributes strain. Section 2 introduces the simulation framework—coarse-grained polymer representation, nanoparticle model and boundary conditions, preparation of microphase separated states, and the uniaxial tensile protocol. Section 3 reports the morphology-dependent mechanical response, including stress–strain behavior and the emergence of a reinforcement-to-weakening crossover, and interprets the strain field distribution in terms of interfacial buffering and connectivity changes; the section closes by discussing implications and limitations.

2. Models and methods

This section defines the computational setting used throughout the study. We outline the coarse-grained mapping of soft and hard segments and the construction of a single nanoparticle embedded at the cell center under periodic boundaries to represent a well-dispersed baseline. The procedure used to generate microphase-separated PU matrices was also described. To verify the potential function used, the uniaxial tensile loading simulation was performed on each homogeneous phase, and the stress–strain response is quantitatively evaluated. Finally, the morphology of the microphase designed by hard segment content was analyzed (see **Fig. S2** for workflow)

2.1. Computational details

In this study, both PU molecules and nanoparticles were described as coarse-grained frameworks. By reducing the molecular details to bead-like units, the coarse-grained PU chains exhibited minimal intermolecular friction. This simplification enabled the efficient observation of phase separation dynamics within the nanosecond time scale, which was well within the temporal resolution of MD simulations [46]. All interaction parameters for the constituent beads were defined by the MARTINI force field [47–49]. The MARTINI framework classified coarse-grained beads into four primary types: polar (P), non-polar (N), apolar (C), and charged (Q). Each bead type was further subcategorized according to its hydrogen-bonding capability (d = donor, a = acceptor, da = both, and 0 = none) and polarity index (assigned as an integer from 1 to 5, depending on the dipole strength) [50]. This parameterization scheme enabled MARTINI-based simulations to reliably capture polymer microphase separation and reorganization behaviors [51]. Moreover, the MARTINI force field had previously been validated across a broad range of material systems and successfully reproduced diverse mechanical responses, ranging from glassy rigidity driven by strong intermolecular interactions to hyperelastic behavior exhibited by soft matter such as elastomers [52].

Coarse-grained modeling and system construction were performed using a commercial MD platform, Materials Studio 2020 (Dassault Systèmes, France). Phase-separation simulations and mechanical-property evaluations were performed using the open-source code, LAMMPS (Sandia Lab) [53].

2.2. Molecular modeling of microphase-separated PUs and their nanocomposites

The procedure for fabricating the microphase-separated PU followed that described in our previous study [52]. Coarse-grained bead modeling of the PU molecules was performed in accordance with the MARTINI standard (Fig. 1**(a)**, see **Fig. S3** for mapping scheme) [53–56]. The chemical composition of the PU was adapted from the literature [57]. Three isophore diisocyanates linearly extended with ethylene glycol were prepared as hard segments. Poly(tetramethylene ether glycol) was considered as a soft segment to bridge the hard domains. To systematically control the hard segment content, the length of the soft segment (i.e., n in Fig. 1**(a)**) varied from 14 to 70 repeat units; thus, the hard segment content ranged from 13.3 to 43.5 wt. %. The total molecular weight of each PU chain was fixed at approximately 30,000 g/mol for all compositions. This design ensured similar characteristic lengths across all PU molecules, eliminating potential variations in the phase behavior or mechanical response arising from differences in molecular size.

A total of 20 PU chains with different hard segment contents were randomly distributed within a cubic simulation box. To prevent artificial overlap or interference during initial placement, the system was initialized at a low density of 0.1 g/cm^3. An initial NVT equilibration was conducted at 300 K for 5 ns using the Verlet integration scheme with a 5 fs time step. In the modeling process, the effects of external gas or moisture inflow were completely excluded. Subsequently, the system gradually densified under NPT conditions (300 K and 1 atm) for 10 ns. After sufficient densification, thermal annealing was performed at 500 K for 10 ns to obtain a homogeneous amorphous structure with complete mixing of the hard and soft segments. Subsequently, the annealed system was subjected to long-term relaxation for 100 ns at 300 K and 1 atm in the NPT ensemble. The potential energy, temperature, and mass density were sufficiently converged during 100 ns dynamics (see **Fig. S4**), indicating that the designed PU has achieved sufficient internal equilibrium. The PU molecules then self-organized into

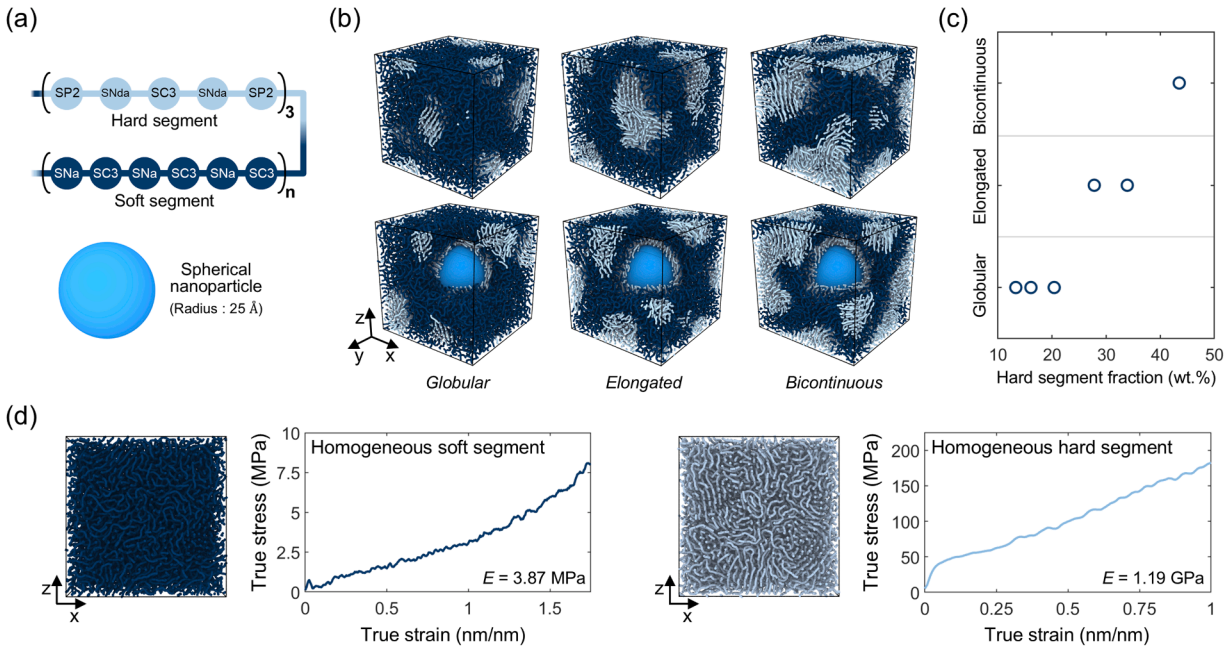


Fig. 1. Preparation of microphase-separated PU and PU nanocomposites. (a) MARTINI-based coarse-grained mapping of the PU molecule and nanoparticle. (b) Molecular configuration of equilibrated PU (top) and PU nanocomposites (bottom). Hard segment content of 20.4, 33.9, and 43.5 wt. % from left to right. (c) Microphase evolution as a function of the hard segment fraction.

thermodynamically favorable microphase-separated structures, allowing the formation of distinct hard and soft domains (Fig. 1**(b)**).

Nanocomposite modeling was performed using the procedure described above, except that single spherical nanoparticles were incorporated. The coarse-grained parameters of the nanoparticles were derived from the study by Hente et al. [58]. They described a boehmite nanofiller, an aluminum oxyhydroxide mineral (γ-AlO(OH)) with OH-terminated surface groups. Interaction parameters derived using the arithmetic mixing rule indicated that the nanoparticles exhibit a stronger energetic affinity for the hard segment compared to the soft segment (40.23 kJ/mol vs. 35.48–37.93 kJ/mol, see **Table S1** and **Table S2**), reflecting the surface characteristics of the OH-terminated boehmite nanofiller. In this study, nanoparticles were modeled as single rigid beads, with their effective radius determined by the van der Waals distance. The effective van der Waals radius of the PU-embedded nanoparticle bead was set to 30 Å, corresponding to a nanoparticle sphere with a diameter of 60 Å. We also verified that the MARTINI-based potential calibrated with arithmetic mixing rule adequately describes the interactions between PU and nanoparticles and the differences in energetic preferences between hard and soft segments (see **Potential verification** section and **Fig. S5** in Supplementary Data).

During the progression of microphase separation, various nanostructure morphologies were observed (Fig. 1**(c)**). The hard segments preferentially aggregated into spherical (i.e., globular) nanodomains that were fully embedded within the continuous soft phase at low hard segment contents (13.3–20.4 wt. %). With an increase in hard segment fraction (27.8–33.9 wt. %), the nanodomains started to interconnect, producing long and interpenetrating (i.e., elongated) structures. At the critical threshold of the hard segment content (43.5 wt. %), the hard phase became fully percolated (i.e., bicontinuous) throughout the simulation cell and extended beyond the periodic boundaries.

Verification of the applied coarse-grained parameters was performed based on the effective mechanical properties of the individual segments. A molecular chain composed of only hard or soft segments was prepared, and a homogeneous-phase microstructure was fabricated according to the identical procedure. After sufficient equilibration, the effective mechanical responses of the hard and soft segments were evaluated by applying uniaxial tensile strain to the equilibrated model. While the models were deformed at a constant true strain rate of 10^8/s, the stress tensor on the constituent atoms was derived using the virial theory [59–61].

The results of the mechanical stress analysis by segment type are presented in Fig. 1**(d)**. The soft segments displayed hyperelastic behavior characterized by a steepening slope in the stress–strain profile with increasing strain. There was no clear elastic–plastic transition, and the stress levels remained remarkably low, not exceeding 10 MPa. The elastic modulus of the homogeneous soft phase obtained by linearly fitting the profile up to a 2.5 % strain was 3.87 MPa. This was very close to a typical estimate of the room temperature modulus of macrodiol of 3 MPa via atomic force microscopy [62]. In contrast, the hard segments exhibited linear strain hardening with a distinct yield point. The elastic modulus of the homogeneous hard phase was found to be 1.19 GPa. Previous studies on the hard domain of urethane-containing polymers reported that the relevant elastic moduli reached 2.5–5 GPa [63,64], similar levels to the current study. These all demonstrated that the data obtained using the MARTINI force field very well predict the differences in mechanical response characteristics for segment types constituting PU. Additionally, the MARTINI-based potential set used successfully estimated the thermal behavior of PU (see **Glass transition temperature evaluation** and **Fig. S6** of Supplementary Data). Experimental validation of the MARTINI force field for commercial PU is presented in **Fig. S7** in the **Experimental Validation** section of Supplementary Data.

3. Results and discussion

In this section, we first establish the baseline microphase morphology and mechanical response of the PU matrix, then quantify how adding a nanoparticle modifies both structure and mechanics across compositions. The section presents mechanical behavior that reveals a morphology-dependent transition from reinforcement to weakening followed by strain field distribution that highlights a role of interfacial layers. Building on these observations, we discuss how microphase state governs the continuity of hard domains and the associated load transfer in PU matrix.

3.1. Mechanical properties under uniaxial tension

Uniaxial tensile deformation simulations were conducted based on the following protocol. A true strain rate of 10^8 s^{-1} was applied along a single axis of the simulation cell. A lateral pressure of 1 atm was maintained along the orthogonal axis to account for Poisson's effect. The temperature was maintained at room condition (i.e., 300 K) throughout all tensile processes, and the internal stress response was computed using the atomic virial stress formulation. Tensile loading continued until the true strain reached 0.9, and the resulting stress–strain response was fitted using the Ogden hyperelastic model [65]. The test was repeated three times along each axis of the unit cell, and the results were averaged. We have demonstrated that even at strain rates much higher than those observed in experiments, converged mechanical behavior can be sufficiently obtained due to technical loading conditions achieved in a simulation environment (see **Strain rate effect** section in Supplementary Data and **Fig. S8**).

The mechanical response of neat PU exhibits a distinct dependence on the microphase morphology (Fig. 2**(a)**). At low hard segment contents (13.3–20.4 wt. %), the hard domains predominantly adopt a globular configuration dispersed within a soft continuous matrix. In this regime, PU displays pronounced hyperelastic behavior with a low elastic modulus. The soft matrix effectively absorbs externally applied mechanical energy owing to its flexibility and acts as a buffering medium surrounding rigid inclusions. Meanwhile, the substantial contribution of the hard segment becomes pronounced only at high strain levels (true strain > 0.6), where the extended chains start activating stress transfer to the hard domains. As the hard segment content increases (27.8–43.5 wt. %), the morphology gradually evolves into anisotropic, elongated structures and bicontinuous structures. As the morphology changes, a continuous load transfer path is promoted throughout the hard domain, significantly increasing elastic stiffness. Accordingly, the mechanical behavior of the entire nanocomposite transitions from hyperelastic to elastoplastic.

The incorporation of nanoparticles significantly altered the microphase-dependent mechanical behavior of PU. At low hard segment concentrations, the nanoparticles contributed to the mechanical reinforcement of the PU materials. Conversely, in environments with high hard segment content, the presence of nanoparticles led to the mechanical degradation of the overall system. Such contradictory effects of nanoparticle incorporation are in good agreement with previous experimental results, highlighting the dependence of the reinforcing capacity of PU-based nanocomposites on their molecular composition [18]. For further quantitative evaluation, the elastic modulus (E) and mechanical toughness (U_T) were calculated from the stress–strain curves (Fig. 2**(b)**). The enhancement factor was then defined as the ratio of the properties of the nanocomposite and neat PU (i.e., E_{neat}/E_{nc} or $U_{T,neat}/U_{T,nc}$ where the subscripts *neat* and *nc* denote the neat PU and nanocomposite systems, respectively) (Fig. 2**(c)**). The quantitative values of the simulation results are tabulated in **Table S3**. The proposed factor shows a clear inverse relationship with the hard segment fraction. The improvement of the mechanical properties owing to nanoparticle incorporation gradually decreases as the hard segment content increases. In particular, the change in the enhancement factor was not linear, but showed a stepwise trend with the morphology transition of PU. This suggests that the reinforcing capacity of nanocomposites are determined by competition between two factors, namely, cohesion of the matrix and interaction with nanoparticles.

3.2. Characterization of nanoparticle-modulated microphase morphology

The spatial distribution of the PU segments was analyzed to elucidate nanoparticle-induced microphase modulation. The simulation cell was partitioned into concentric spherical shells with a radial thickness of 1 Å, centered at the nanoparticle. The local concentration of hard segments within each shell was then computed individually (Fig. 3**(a)** and **(b)**). The analysis reveals three distinct morphological transitions induced by nanoparticles. First, owing to preferential energetic interactions, the region surrounding the nanoparticles is predominantly enriched with hard segments. This selective adsorption arises from the favorable thermodynamic affinity between the hard segments and nanofiller. Previous experimental approaches using FT-IR analysis suggested abundant hydrogen bonds between the nanoparticles and the urethane groups [66]; this further supports the selective adsorption of hard segments on the nanoparticle surface. Second, the next region beyond the interfacial layer exhibits a hard segment content reduction. This is due to the structural constraints of the soft segments, which are excluded from interfacial adsorption while being interconnected with the hard segments. The depletion region was calculated according to the van der Waals criteria of the hard segment, and the thickness of the region was observed to be 24.3 Å in average. Third, the influence of the filler diminishes at greater distances from the nanoparticles, and the distribution of the hard and soft segments gradually returns to a homogeneous state. Consequently, the hard segments adsorbed onto the nanoparticles become spatially isolated, inhibiting the formation of a continuous, rigid network. In other words, the nanoparticles act as physical barriers that confine hard segments, forming "hard segment islands" within the PU microstructure.

Analysis of the microphase using atomic force microscopy suggests that insertion of nanoparticles promotes the progression of phase

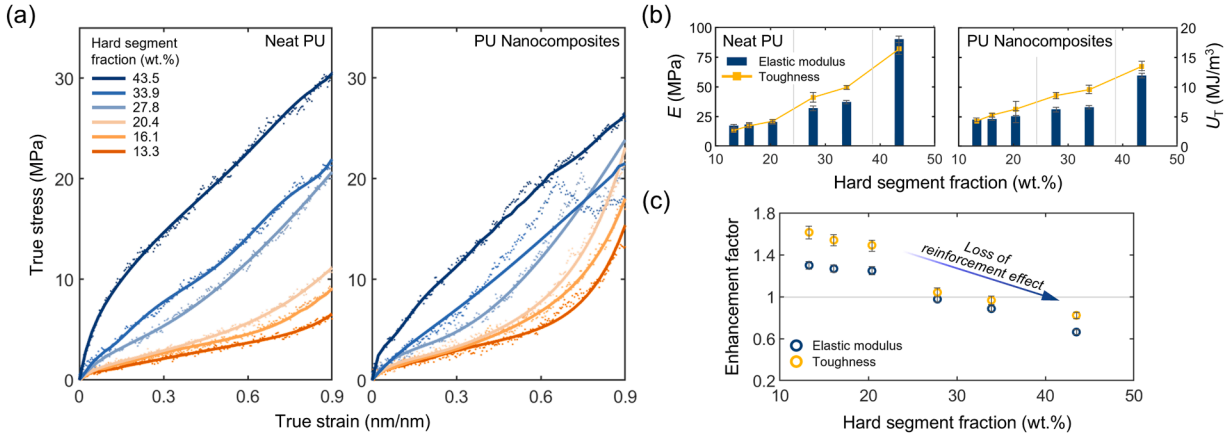


Fig. 2. Mechanical property characterization of PU and PU nanocomposites. (a) Stress–strain profile of neat PU (left) and PU nanocomposites (right) as a function of hard segment fraction. The dotted line represents the raw profile, and the solid line represents the hyperelastic fitting using Ogden model. (b) Variations in elastic modulus (E), toughness (U_T), and (c) their enhancement factor relative to the hard segment content.

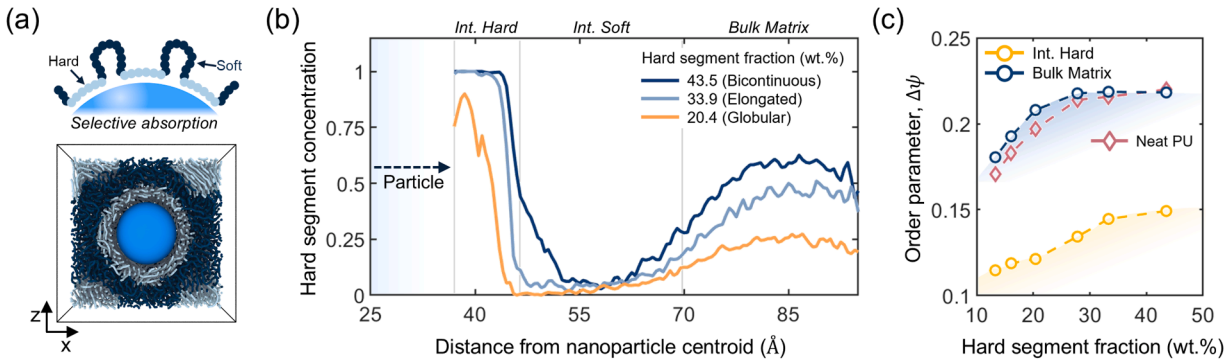


Fig. 3. Characterization of microphase morphology influenced by nanoparticle incorporation. (a) Cross-sectional view of the nanocomposite model with 43.5 wt. % hard segment fraction. Nanoparticles explicitly exhibit selective adsorption to hard segments. (b) Radial distribution of hard segments from the center of the nanoparticle. The closer it is to 1, the more the area is completely filled with hard segments, and vice versa. (c) Order parameter of hard segments ($\Delta\psi$) located in the interfacial region and bulk matrix.

separation and the distinctness of each phase [40–42]. This supports current observations that nanoparticles act as nucleation site for phase separation.

The connectivity of the hard domains according to spatial location was further analyzed for hard segments located at the interface and in the bulk matrix. The order parameter ($\Delta\psi$) was calculated using the following equation [67–69]:

$$\Delta\psi = \frac{1}{N}\sum_{i=1}^{N}\frac{(N_{HS}-N_{SS})2}{(N_{HS}+N_{SS})2}, \tag{1}$$

where N_{HS} and N_{SS} denote the number of hard and soft segments within 0.75 nm of the i th bead. From a structural perspective, higher parameter values indicate that the bead is more involved in the hard domain, whereas lower values indicate that the bead is more involved in the soft domain.

The order parameter of the neat PUs exhibits a monotonic increase with increasing hard segment content (Fig. 3**(c)**). This is intuitive because an increasing hard segment concentration leads to the formation of a globular hard domain with an increasing diameter. In addition, as the domain size increases, the surface area-to-volume ratio of the aggregates decreases, thereby reducing the interfacial area available for interactions with the surrounding soft segments. Consequently, the order parameter values progressively saturates as the hard domain undergoes morphological evolution (i.e., globular–elongated–bicontinuous).

In nanocomposite systems incorporating nanoparticles, the order parameter value of the bulk hard domains remains comparable to that of neat PU systems. However, significantly lower-order parameter values (< 0.15) are consistently observed for hard domains localized near the nanoparticle surface, regardless of the overall hard segment content. The suppression of local connectivity is attributed to the spatial isolation of hard segments at the nanoparticle interface. This finding is in good agreement with the observation that preferential and selective adsorption onto the filler surface spatially impedes the percolation of the rigid network.

To assess the spatially dependent mechanical response of hard segments, virtual peel tests were performed. The representative microstructures of the hard domains were separately obtained from both the bulk matrix and nanoparticle interfacial region (Fig. 4**(a)**). A hard segment molecule adsorbed at the domain surface was then subjected to mechanical displacement by applying a constant velocity of 0.05 Å/ps to one terminal atom. During the controlled separation process, the tensile force acting on the segment was recorded as a function of displacement. Such simulation-derived peel test enables ideal conditions for independent assessment of interfacial properties. In other words, the virtual peel test minimizes external factors that could potentially influence the results. In experimental approaches including tensile test or pull-out test, the mechanical load applied to the interfacial area must be transmitted by the surrounding matrix, making it very challenging to isolate external factors. This is particularly important for the present system, as the results of the present study clearly demonstrate that nanoparticles exert long-range effects through hierarchical structures from the interface (interfacial hard rich–subsequent hard depletion, see Fig. 3). In contrast, the virtual peel test directly imposes displacements on chains located at the interface or in the bulk crystalline and explicitly extracts the traction force on the displacing chains with nanoscale resolution, thereby contributing to a rigorous assessment of structural rigidity.

The results clearly demonstrate that the hard segments residing at the nanoparticle interface exhibit enhanced resistance to mechanical detachment compared to those in the bulk matrix (Fig. 4**(b)**). The force–displacement curves reveal that the interfacial domain maintains a consistently higher peeling force throughout the entire separation process. Specifically, the interfacial peel force converges at

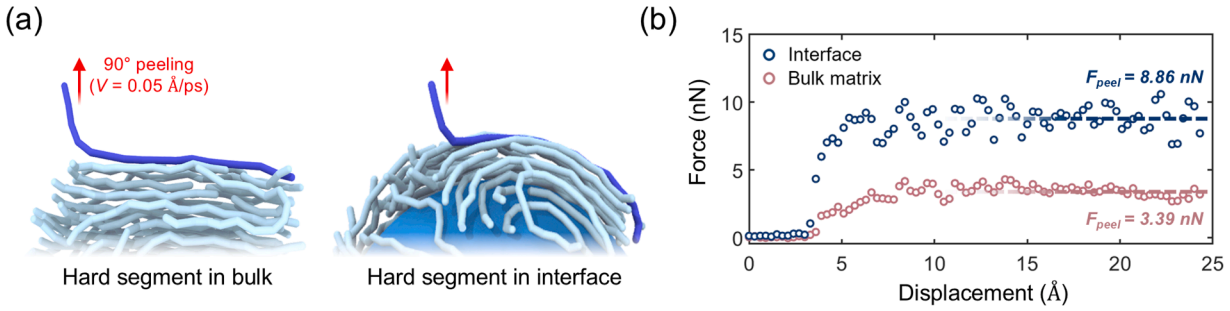


Fig. 4. Mechanical strength measurements of the hard domain. (a) Snapshots of the 90° peel test simulation of the bulk and interfacial hard segments. Purple indicates the arbitrary chain selected for measurement. (b) Peel forces of the bulk and interfacial hard segments as a function of vertical displacement. A higher structural rigidity is observed at the interfacial hard segment than the bulk crystalline.

approximately 8.86 nN, which is 2.62 times greater than the bulk counterpart value of 3.39 nN. This pronounced reinforcement is attributed to strong physical confinement by the nanoparticle surface, which restricts molecular mobility and elevated local cohesion.

Based on the above findings, the effects of nanoparticles on the microstructure of PU can be summarized as follows (Fig. 5). The nanoparticles generate hard domains that are confined at their interfaces. The hard segments adsorbed at the interface exhibit enhanced structural integrity compared with those in the bulk region. The effect of such microphase modulation by nanoparticles varies, particularly depending on the hard segment content. In globular PU, the incorporation of nanoparticles leads to morphological consistency of the microphase. The physical adsorption of nanoparticles further enhances load-carrying efficiency by serving as enhanced physical cross-linking sites. Conversely, in elongated and bicontinuous PUs, load transfer predominately occurs through a rigid network composed of interconnected hard domains. Confined hard domains derived from nanoparticles reduce the connectivity of the rigid network, leading to structural deterioration. Consequently, the reinforcing efficiency of nanoparticles is largely governed by the composition of PU molecules and the associated microphase morphology.

3.3. Demonstration of microphase-dependent mechanical role of nanoparticles

Sub-continuum stress decomposition was performed to further characterize the contrasting mechanical effects of nanoparticles in PU nanocomposites. The virial stress tensor captures the stress states of a system by incorporating the thermal kinetics and interatomic potential energies of its constituents. By refining the energy states of individual constituents, virial calculations enable the spatial partitioning of stress contributions. Atomic-wise, domain-specific, or field-level mechanical evaluations can thus be achieved based on virial formulation [70–72]. The PU nanocomposites were divided into three regions, namely, nanoparticles and adsorbed hard segments (NP + Interfacial HS), hard segments located in the bulk matrix (Bulk HS), and soft segment matrix (SS). During the application of the uniaxial tension, the stress tensors that evolved in each region were derived individually. The mechanical contributions of each domain were then quantified using the strain energy density. The theoretical foundation of this approach has been established in our previous studies [70].

The strain energy density analysis reveals differential reinforcement mechanisms governed by nanoconfinement and phase separation. (Fig. 6). The SS domain exhibits a relatively consistent strain energy profile across various hard segment contents. This is reasonable, considering that the SS consistently maintained a continuous and compliant network for the hard segment fractions considered. In contrast, the Bulk HS domain demonstrates a pronounced increase in strain energy density with increasing hard segment fraction. This is attributed to the progressive formation of percolating hard domains, which establish continuous load transfer pathways through microphase coalescence.

Notably, in systems with a globular hard domain morphology, the NP + Interfacial HS domain shows a significantly elevated strain energy contribution compared to the Bulk HS region. This enhancement becomes particularly apparent under large tensile strains (true strain > 0.6), where chain alignment and segmental extension activate intramolecular load transfer mechanisms. However, as the hard segment fraction produces a transition from isolated globules to an interconnected hard domain network, the relative contribution of the NP + Interfacial HS domain diminishes. Such a transition suggests that the fragmentation of the continuous rigid phase dominates the mechanical contribution of the nanoparticles.

These spatially heterogeneous transformations also explicitly account for their role in fatigue and cyclic loading. At low hard segment contents, nanoparticle can enhance reversible load sharing and delay microdamage at small–moderate cyclic strains. When the hard phase is already continuous, nanoparticle interfaces create hard segment islands that suppress local connectivity. Under cyclic loading, such interfacial partitions are expected to serve as stress-redistribution sites, promoting earlier interfacial debonding/cavitation, and shorter fatigue life.

The molecular origin of the observed contrasting effect was further revealed by the von Mises strain distribution (Fig. 7) [73–75]. Owing to differences in inherent stiffness, the soft domain consistently exhibits higher local strain intensities. From a structural perspective, the soft segment layer surrounding the nanoparticles and interfacial hard domains acts as a compliant interphase (i.e., buffer layer). The region effectively absorbs mechanical energy through active molecular rearrangement and segmental mobility in response to external deformation. This structural transition is critical in environments with high concentrations of rigid segments, where a continuous load-bearing network becomes key in mechanical responses. Ultimately, the nanoparticles adapt to the nanoscale mechanical integrity, inducing contrasting effects on the macroscopic mechanical properties.

4. Conclusion

In this study, the molecular mechanisms governing the reinforcement efficiency of PU nanocomposites were systematically investigated via atomistic simulations. The results highlight that nanoparticle efficacy is strongly modulated by interfacial interactions and the associated phase-specific spatial distribution. The unique insights derived from this study can be summarized in the following two points. (1) Effective percolated threshold of hard segments for microphase evolution was determined: The microstructure of PU was dominated by the content of hard segments and showed a stepwise evolution of globular–elongated–bicontinuous. From the lowest content (13.3 wt. %), the critical threshold for structural evolution is 27.8 wt. % for elongated and 43.5 wt. % for bicontinuous. It should be emphasized that previous experimental approaches using AFM phase images or TEM have been challenging in accurately identifying such microstructure or microphase morphology due to their inherent resolution. This also explains the

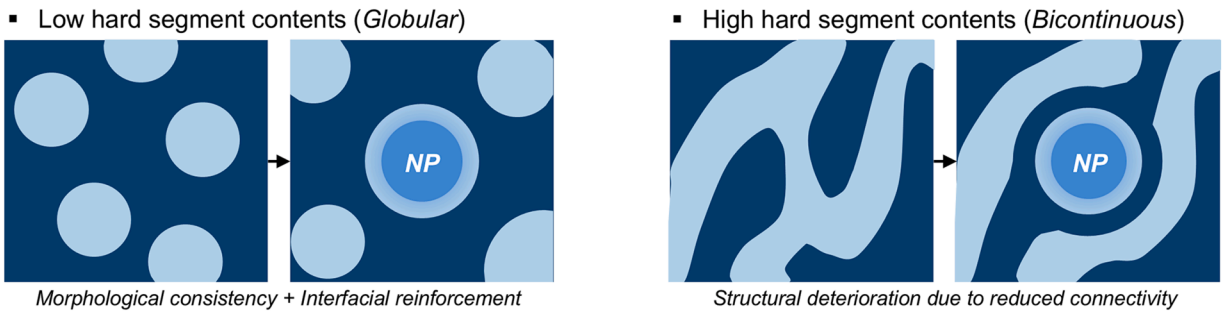


Fig. 5. Schematic of the nanoparticle-driven microphase transition in PU. In all hard segment fractions, nanoparticles induce identical morphology transitions, i.e., hard segment islands, but their influence varies depending on the microphase morphology.

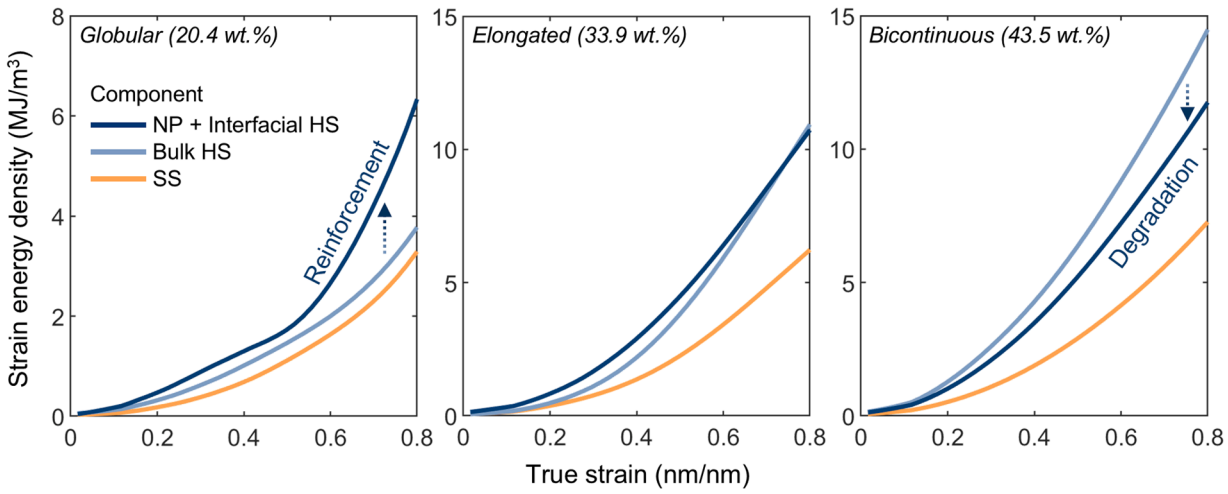


Fig. 6. Spatial distribution of strain energy density in PU nanocomposites under mechanical tension. Higher values are interpreted as greater resistance to mechanical loads.

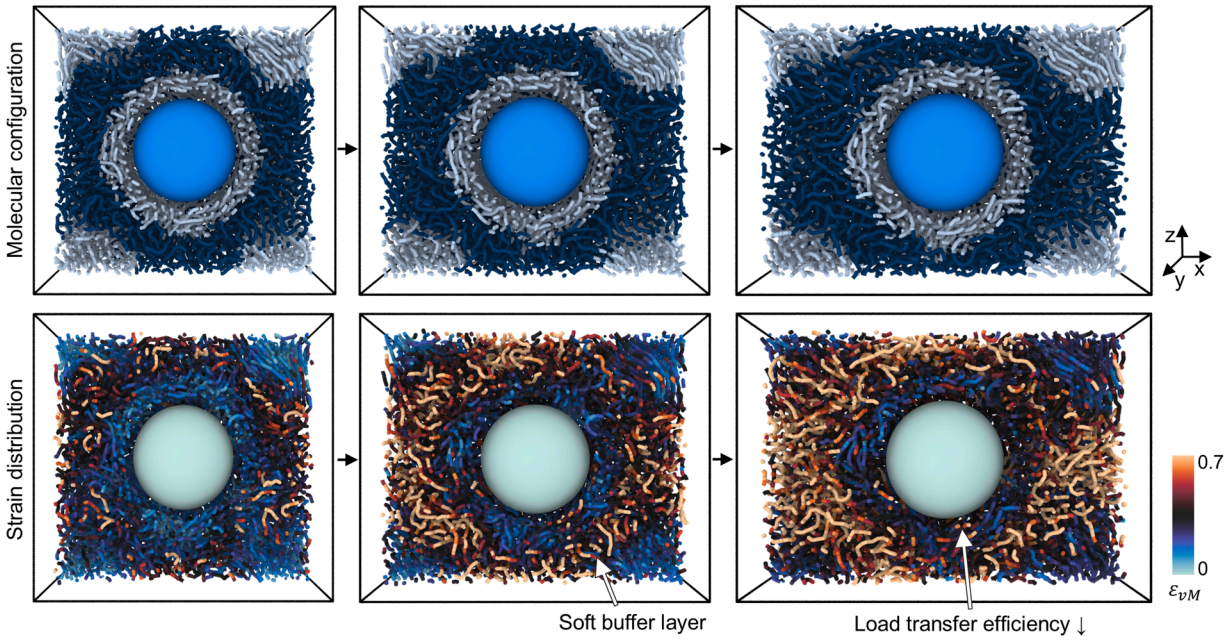


Fig. 7. Molecular configuration (top) and von Mises strain distribution (ε_{vM}, bottom) of 44.3 wt. % PU nanocomposites during uniaxial deformation. Applied true strain of 0.01, 0.2, and 0.4 from left to right.

stepwise and nonlinear property enhancement of PU materials with hard segment fraction that experimentally observed. (2) Hard segment fraction is quantitatively determined to rationally use the reinforcing effect. The industrial applications of PU range from energy-absorbing materials based on soft and hyperelastic behavior to load-bearing components based on rigid and glassy behavior. To achieve each purpose, nanoparticles must be used with careful consideration of the hard segment fraction. The reinforcing effect is effective only at hard segment contents below 20.4 wt. %, and the point where mechanical properties are substantially enhanced is dominant at high strain levels. On the other hand, nanoparticles at high hard segment contents of 43.5 wt. % can be a strategy to make the material flexible and deformable. In addition, in the range between 27.8 and 33.9 wt. % for elongated morphology, the influence of mechanical properties by nanoparticles is negligible; so, the insertion of nanoparticles only has the negative effect of increasing the complexity and cost of the manufacturing process. It is also novel point revealed for the first time that the change in reinforcing efficiency is quite insensitive to the hard segment content for the identical morphology. Moreover, the mechanism of the above two phenomenon was adequately supported by subcontinuum level microstructural analysis: hard segment island in nanoparticle surfaces.

The findings demonstrate that the mechanical role of nanoparticles is microphase-dependent and that their incorporation must be carefully tailored according to the morphological state of the PU matrix. In this regard, this study provides theoretical guidelines for optimizing reinforcement for PU-based nanocomposites through microphase engineering. The sub-continuum microanalysis provides a clear, physics-based interpretation of mechanical work and strain-field distributions. Together, these insights provide a strategy to deliberately tune interfacial load transfer and local strain concentration within separated microphases, enabling the design of next-generation energy-absorbing/dissipative materials and wear-resistant coatings [76,77].

This study examines the well-dispersed baseline in which nanoparticles are uniformly spaced, enabling a clean assessment of interfacial

adsorption and hard-network connectivity under tensile loading. Practical systems introduce added complexity through size, shape anisotropy, and aggregation, which modify surface-to-volume ratio, interfacial coverage, and network partitioning [78–80]. Manually controlling the affinity between nanoparticles and PU through surface chemistry induces additional modulation of phase separation behavior. The design variable of PU molecules, including sequence distribution and hard segment length, also governs the critical percolation threshold of hard domain connectivity. In this regard, the insights of this study can serve as a theoretical foundation for microphase-dependent mechanical effect. Future simulations will expand to practical nanocomposite parameters with controlled molecular composition, filler volume fraction, aspect ratio, and cluster statistics to quantify how these parameters shift the predicted reinforcement–weakening crossover and to derive morphology-aware design guidelines.

CRediT authorship contribution statement

Hongdeok Kim: Writing – original draft, Visualization, Software, Methodology, Data curation. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmecsci.2025.111023.

Data availability

Data will be made available on request.

References

[1] Kotanen S, Laurikainen P, Lehtimäki S, Harjunalanen T, Laaksonen T, Sarlin E. The role of hard and soft segments in the thermal and mechanical properties of non-isocyanate polyurethanes produced via polycondensation reaction. Int J Adhes Adhes 2024;132:103726. https://doi.org/10.1016/j.ijadhadh.2024.103726.
[2] Yu W, Du M, Zhang D, Lin Y, Zheng Q. Influence of dangling chains on molecular dynamics of polyurethanes. Macromolecules 2013;46(18):7341–51. https://doi.org/10.1021/ma401260d.
[3] Ma R, Wang Y, Huang H, Zhao X, Li X, Zhang L, Gao Y. Insight into the fracture properties and molecular mechanism of polyurethane. Macromolecules 2024;57(9):4484–94. https://doi.org/10.1021/acs.macromol.4c00346.
[4] Koberstein JT, Galambos AF, Leung LM. Compression-molded polyurethane block copolymers. 1. Microdomain morphology and thermomechanical properties. Macromolecules 1992;25(23):6195–204. https://doi.org/10.1021/ma00049a017.
[5] Sánchez-Adsuar MS, Pastor-Blas MM, Martín-Martínez JM. Properties of polyurethane elastomers with different hard/soft segment ratio. J Adhes 1998;67(1–4):327–45. https://doi.org/10.1080/00218469808011115.
[6] Zhu A, Shi L, Shi H, Qian J, Shi Y. Role of the soft and hard segments structure in modifying the performance of acrylic adhesives modified by polyurethane macromonomers. ACS Omega 2024;9(30):32735–44. https://doi.org/10.1021/acsomega.4c02710.
[7] Korley LTJ, Pate BD, Thomas EL, Hammond PT. Effect of the degree of soft and hard segment ordering on the morphology and mechanical behavior of semicrystalline segmented polyurethanes. Polymer (Guildf) 2006;47(9):3073–82. https://doi.org/10.1016/j.polymer.2006.02.093.
[8] Koberstein JT, Galambos AF. Multiple melting in segmented polyurethane block copolymers. Macromolecules 1992;25(21):5618–24. https://doi.org/10.1021/ma00047a010.
[9] Velankar S, Cooper SL. Microphase separation and rheological properties of polyurethane melts. 2. Effect of block incompatibility on the microstructure. Macromolecules 2000;33(2):382–94. https://doi.org/10.1021/ma990817g.
[10] Velankar S, Cooper SL. Microphase separation and rheological properties of polyurethane melts. 3. Effect of block incompatibility on the viscoelastic properties. Macromolecules 2000;33(2):395–403. https://doi.org/10.1021/ma9908189.
[11] Cheng BX, Gao WC, Ren XM, Ouyang XY, Zhao Y, Zhao H, Wu W, Huang CX, Liu Y, Liu XY, Li HN, Li RK. A review of microphase separation of polyurethane: characterization and applications. Polym Test 2022;107:107489. https://doi.org/10.1016/j.polymertesting.2022.107489.
[12] Zhang Q, Lin X, Chen W, Jiang K, Han D. Applications of characterization methods in polyurethane materials: analysis of microphase-separated structures. Appl Spectrosc Rev 2022;57(2):153–76. https://doi.org/10.1080/05704928.2021.1975126.
[13] Liu B, Niu Z, Wang Z, Wang Y, Zou H, Zhao X, Hu S. Effect of hard segment content on the phase separation and properties of hydroxyl-terminated polybutadiene thermoplastic polyurethane. Polym Test 2025;143:108681. https://doi.org/10.1016/j.polymertesting.2024.108681.
[14] Tian Y, Wei Y, Wang M, Wang J, Li X, Qin X, Zhang L. Ultra-stretchable, tough, and self-healing polyurethane with tunable microphase separation for flexible wearable electronics. Nano Energy 2025;139:110908. https://doi.org/10.1016/j.nanoen.2025.110908.
[15] Desper CR, Schneider NS, Jasinski JP, Lin JS. Deformation of microphase structures in segmented polyurethanes. Macromolecules 1985;18(12):2755–61. https://doi.org/10.1021/ma00154a067.
[16] Chu B, Gao T, Li Y, Wang J, Desper CR, Byrne CA. Microphase separation kinetics in segmented polyurethanes: effects of soft segment length and structure. Macromolecules 1992;25(21):5724–9. https://doi.org/10.1021/ma00047a025.
[17] Kojio K, Nakashima S, Furukawa M. Microphase-separated structure and mechanical properties of norbornane diisocyanate-based polyurethanes. Polymer (Guildf) 2007;48(4):997–1004. https://doi.org/10.1016/j.polymer.2006.12.057.
[18] Zhou H, Shi Z, Zhou X, Li X, Wen Y, Xue Z, Wang Y, Xie X. Reversible and ultrastrong epoxy-polyurethane adhesives with microphase separation. Adv Funct Mater 2025:2506419. https://doi.org/10.1002/adfm.202506419.
[19] Shi K, Li X, Lei L, Chen P, Ge M, Wu T, Jiang H, Zou M. Influence of dangling chains on the microphase separation and damping properties of polyurethane. J Appl Polym Sci 2024;141(34):e55867. https://doi.org/10.1002/app.55867.
[20] Huang X, Wang G, Zhang Y, Zhou Z, Yu Y. Controlled microphase separation in polyurethane acrylate-based optically clear pressure sensitive adhesives. Prog Org Coatings 2025;200:109064. https://doi.org/10.1016/j.porgcoat.2025.109064.
[21] Qi HJ, Boyce MC. Stress–strain behavior of thermoplastic polyurethanes. Mech Mater 2005;37(8):817–39. https://doi.org/10.1016/j.mechmat.2004.08.001.
[22] Petrović ZS, Ferguson J. Polyurethane elastomers. Prog Polym Sci 1991;16(5):695–836. https://doi.org/10.1016/0079-6700(91)90011-9.
[23] Estes GM, Seymour RW, Cooper SL. Infrared studies of segmented polyurethane elastomers. II. Infrared dichroism. Macromolecules 1971;4(4):452–7. https://doi.org/10.1021/ma60022a018.
[24] Huang M, Li Z, Xu W, He S, Liu W, Jiang L, Liu H. Visualization and quantification of microphase separation in thermoplastic polyurethanes under different hard segment contents and its effect on the mechanical properties. Polym Test 2024;131:108329. https://doi.org/10.1016/j.polymertesting.2024.108329.
[25] Garrett JT, Runt J, Lin JS. Microphase separation of segmented poly(urethane urea) block copolymers. Macromolecules 2000;33(17):6353–9. https://doi.org/10.1021/ma000600i.
[26] Erekkath S, Sreejalekshmi KG. Theoretical predictions on microphase separation in polyurethane: combinatorial design, synthesis and demonstration of shape memory property. Mater Today Commun 2018;16:71–80. https://doi.org/10.1016/j.mtcomm.2018.04.007.
[27] Jung YS, Lee S, Park J, Shin EJ. One-shot synthesis of thermoplastic polyurethane based on bio-polyol (polytrimethylene ether glycol) and characterization of microphase separation. Polymers (Basel) 2022;14(20):4269. https://doi.org/10.3390/polym14204269.
[28] Jiang K, Chen W, Liu X, Wang Y, Han D, Zhang Q. Effect of bio-based polyols and chain extender on the microphase separation structure, mechanical properties and morphology of rigid polyurethane foams. Eur Polym J 2022;179:111572. https://doi.org/10.1016/j.eurpolymj.2022.111572.
[29] Liu S, Wang Y, Song ZQ, Liu YD, Yang J, Lu G, Liang Y. Relationship between multiscale structure and mechanical hysteresis loss in thermoplastic polyurethane-urea elastomer films. J Appl Polym Sci 2025:e57556. https://doi.org/10.1002/app.57556.
[30] Akindoyo JO, Beg M, Ghazali S, Islam MR, Jeyaratnam N, Yuvaraj AR. Polyurethane types, synthesis and applications–a review. RSC Adv 2016;6(115):114453–82. https://doi.org/10.1039/C6RA14525F.
[31] Somarathna HMCC, Raman SN, Mohotti D, Mutalib AA, Badri KH. The use of polyurethane for structural and infrastructural engineering applications: a state-of-the-art review. Constr Build Mater 2018;190:995–1014. https://doi.org/10.1016/j.conbuildmat.2018.09.166.
[32] Joseph J, Patel RM, Wenham A, Smith JR. Biomedical applications of polyurethane materials and coatings. Trans IMF 2018;96(3):121–9. https://doi.org/10.1080/00202967.2018.1450209.
[33] Naureen B, Haseeb ASMA, Basirun WJ, Muhamad F. Recent advances in tissue engineering scaffolds based on polyurethane and modified polyurethane. Mater Sci Eng 2021;118:111228. https://doi.org/10.1016/j.msec.2020.111228.
[34] Vervaeke R, Debruyne S, Vandepitte D. Numerical and experimental analysis of vibration damping performance of polyurethane adhesive in machine operations.

Int J Adhes Adhes 2019;90:47–54. https://doi.org/10.1016/j.ijadhadh.2019.01.026.
[35] Verdejo R, Stämpfli R, Alvarez-Lainez M, Mourad S, Rodriguez-Perez MA, Brühwiler PA, Shaffer M. Enhanced acoustic damping in flexible polyurethane foams filled with carbon nanotubes. Compos Sci Technol 2009;69(10):1564–9. https://doi.org/10.1016/j.compscitech.2008.07.003.
[36] Finnigan B, Martin D, Halley P, Truss R, Campbell K. Morphology and properties of thermoplastic polyurethane nanocomposites incorporating hydrophilic layered silicates. Polymer (Guildf) 2004;45(7):2249–60. https://doi.org/10.1016/j.polymer.2004.01.049.
[37] Bistričić L, Baranović G, Leskovac M, Bajsić EG. Hydrogen bonding and mechanical properties of thin films of polyether-based polyurethane–silica nanocomposites. Eur Polym J 2010;46(10):1975–87. https://doi.org/10.1016/j.eurpolymj.2010.08.001.
[38] Fortunati E, Luzi F, Janke A, Häußler L, Pionteck J, Kenny JM, Torre L. Reinforcement effect of cellulose nanocrystals in thermoplastic polyurethane matrices characterized by different soft/hard segment ratio. Polym Eng Sci 2017;57(6):521–30. https://doi.org/10.1002/pen.24532.
[39] Chen CJ, Tsai MH, Tseng IH, Hsu AW, Liu TC, Huang SL. Composition, thermal and tensile properties of polyurethane-urea-silica hybrids. RSC Adv 2013;3(25):9729–38. https://doi.org/10.1039/C3RA23186K.
[40] Xia HS, Song M, Zhang ZY, Richardson M. Microphase separation, stress relaxation, and creep behavior of polyurethane nanocomposites. J Appl Polym Sci 2007;103(12):2992–3002. https://doi.org/10.1002/app.25462.
[41] Mohammadzadeh F, Haddadi Asl V, Balzade Z. Sahebi Jouibari I. Switch segment and halloysite nanotube role in the phase separation behavior of shape memory thermoplastic polyurethane. Polym Compos 2020;41(7):2625–33. https://doi.org/10.1002/pc.25561.
[42] Phua SL, Yang L, Toh CL, Huang S, Tsakadze Z, Lau SK, Mai YW, Lu X. Reinforcement of polyether polyurethane with dopamine modified clay: the role of interfacial hydrogen bonding. ACS Appl Mater Interfaces 2012;4(9):4571–8. https://doi.org/10.1021/am300947b.
[43] Dora TL, Singh SK, Mishra RR, Yu H, Rawat NK, Verma A. Exploring deformation mechanisms in a refractory high entropy alloy (MoNbTaW). Int J Mech Sci 2025;288:110000. https://doi.org/10.1016/j.ijmecsci.2025.110000.
[44] Won D, Cho H, Kim H, Lee G, Kwon J, Kim J, Hong S, Choi J, Kim SW, Ko SH. Digital selective reversible phase control of monolithically integrated heterogeneous piezoelectric polymer for frequency dependent unimorph. Adv Opt Mater 2022;10(24):2201206. https://doi.org/10.1002/adom.202201206.
[45] Kim H, Choi J. Molecular-scale mechanics of a crystalline polyethylene interface formed by projection of carbon nanotube structures. Surf Interfaces 2024;47:104213. https://doi.org/10.1016/j.surfin.2024.104213.
[46] Kim H, Choi J. A lamellar morphology based computational modeling for predicting the thermal conductivity of semicrystalline polymers. Int J Mech Sci 2024;282:109622. https://doi.org/10.1016/j.ijmecsci.2024.109622.
[47] Marrink SJ, Risselada H, Yefimov S, Tieleman DP, de Vries AH. The MARTINI Force field: coarse grained model for biomolecular simulations. J Phys Chem B 2007;111(27):7812–24. https://doi.org/10.1021/jp071097f.
[48] Monticelli L, Kandasamy SK, Periole X, Larson RG, Tieleman DP, Marrink SJ. The MARTINI coarse-grained force field: extension to proteins. J Chem Theory Comput 2008;4(5):819–34. https://doi.org/10.1021/ct700324x.
[49] Periole X, Marrink SJ. The MARTINI coarse-grained force field. In: Biomolecular simulations: methods and protocols Methods Mol Biol 2012;924:533–65. https://doi.org/10.1007/978-1-62703-017-5_20.
[50] Kim H, Won D, Ko SH, Choi J. Thermodynamic mechanism governing the coalescence of conductive particles in PEDOT:PSS under laser irradiation. Macromolecules 2024;57(5):2048–56. https://doi.org/10.1021/acs.macromol.3c01870.
[51] Vakili H, Mohseni M, Makki H, Yahyaei H, Ghanbari H, González A, Irusta L. Self-assembly of a patterned hydrophobic-hydrophilic surface by soft segment microphase separation in a segmented polyurethane: combined experimental study and molecular dynamics simulation. Polymer (Guildf) 2020;195:122424. https://doi.org/10.1016/j.polymer.2020.122424.
[52] Kim H, Choi J. Molecular scale investigation of the microphase dependent load transfer capability of polyurethane. Macromolecules 2024;57(22):10745–53. https://doi.org/10.1021/acs.macromol.4c01773.
[53] Thompson AP, Aktulga HM, Berger R, Bolintineanu DS, Brown WM, Crozier PS, in 't Veld PJ, Kohlmeyer A, Moore SG, Nguyen TD, Shan R, Stevens MJ, Tranchida J, Trott C, Plimpton SJ. LAMMPS – a flexible simulation tool for particle based materials modeling at the atomic, meso, and continuum scales. Comput Phys Commun 2022;271:108171. https://doi.org/10.1016/j.cpc.2021.108171.
[54] Ghermezcheshme H, Makki H, Mohseni M, Ebrahimi M. MARTINI-based simulation method for step-growth polymerization and its analysis by size exclusion characterization: a case study of cross-linked polyurethane. Phys Chem Chem Phys 2019;21(38):21603–14. https://doi.org/10.1039/C9CP03407B.
[55] Kizilkaya D, Ghermezcheshme H, Sabzevar SE, Makki H, Kacar G. Modeling surface segregation of smart PU coatings at hydrophilic and hydrophobic interfaces via coarse-grained molecular dynamics and mesoscopic simulations. Prog Org Coat 2023;174:107279. https://doi.org/10.1016/j.porgcoat.2022.107279.
[56] Ghermezcheshme H, Makki H, Mohseni M, Ebrahimi M. Hydrophilic dangling chain interfacial segregation in polyurethane networks at aqueous interfaces and its underlying mechanisms: molecular dynamics simulations. Phys Chem Chem Phys 2020;22(45):26351–63. https://doi.org/10.1039/D0CP04244G.
[57] Kwon MS, Yu C, Ha J-K, Ahn J, Lee J, Choi J, Chang T, Min SK. Defect-free acrylic polymers with a near-Poisson distribution prepared via catalyst-free visible-light-driven radical polymerization. ChemRxiv 2023. https://doi.org/10.26434/chemrxiv-2023-dl842.
[58] Hente A, Arash B, Rolfes R. Optimization assisted coarse grained modeling of agglomerated nanoparticle reinforced thermosetting polymers. Polymer (Guildf) 2021;225:123741. https://doi.org/10.1016/j.polymer.2021.123741.
[59] Subramaniyan AK, Sun CT. Continuum interpretation of virial stress in molecular simulations. Int J Solids Struct 2008;45(14–15):4340–6. https://doi.org/10.1016/j.ijsolstr.2008.03.016.
[60] Tsai DH. The virial theorem and stress calculation in molecular dynamics. J Chem Phys 1979;70(3):1375–82. https://doi.org/10.1063/1.437577.
[61] Pendás AM. Stress, virial, and pressure in the theory of atoms in molecules. J Chem Phys 2002;117(3):965–79. https://doi.org/10.1063/1.1484385.
[62] Rohm K, Manas-Zloczower I. A micromechanical approach to TPU mechanical properties: framework and experimental validation. Mech Mater 2023;180:104627. https://doi.org/10.1016/j.mechmat.2023.104627.
[63] Sui T, Salvati E, Ying S, Sun G, Dolbnya IP, Dragnevski K, Prisacariu C, Korsunsky AM. Strain softening of nano-scale fuzzy interfaces causes Mullins effect in thermoplastic polyurethane. Sci Rep 2017;7(1):916. https://doi.org/10.1038/s41598-017-00904-3.
[64] Stanford JL, Young RJ, Day RJ. Formation and properties of urethane-diacetylene segmented block copolymers. Polymer (Guildf) 1991;32(9):1713–25. https://doi.org/10.1016/0032-3861(91)90410-K.
[65] Kim B, Lee SB, Lee J, Cho S, Park H, Yeom S, Park SH. A comparison among Neo-Hookean model, Mooney-Rivlin model, and Ogden model for chloroprene rubber. Int J Precis Eng Manuf 2012;13(5):759–64. https://doi.org/10.1007/s12541-012-0099-y.
[66] Yu F, Huang HX. Simultaneously toughening and reinforcing poly(lactic acid)/thermoplastic polyurethane blend via enhancing interfacial adhesion by hydrophobic silica nanoparticles. Polym Test 2015;45:107–13. https://doi.org/10.1016/j.polymertesting.2015.06.001.
[67] Rex M, Löwen H. Influence of hydrodynamic interactions on lane formation in oppositely charged driven colloids. Eur Phys J E 2008;26:143–50. https://doi.org/10.1140/epje/i2007-10274-4.
[68] Dutta S, Chakrabarti J. Transient dynamical responses of a charged binary colloid in an electric field. Soft Matter 2018;14(22):4477–82. https://doi.org/10.1039/C7SM02552A.
[69] Wang Y, Ma R, Li H, Hu S, Gao Y, Liu L, Zhao X, Zhang L. Effect of the content and strength of hard segment on the viscoelasticity of the polyurethane elastomer: insights from molecular dynamics simulation. Soft Matter 2022;18(21):4090–101. https://doi.org/10.1039/D2SM00463A.
[70] Kim H, Choi J. Subcontinuum interpretation of mechanical behavior for cross-linked epoxy networks. Macromolecules 2022;55(14):5916–25. https://doi.org/10.1021/acs.macromol.2c00593.
[71] Kim H, Choi J. Influence of crystalline structure on creep resistance capability in semi crystalline polymers: a coarse grained molecular dynamics study. Int J Fatigue 2024;188:108517. https://doi.org/10.1016/j.ijfatigue.2024.108517.
[72] Kim H, Choi J. A computational method for characterizing molecular scale load transfer in polymer systems with structural heterogeneity. Polymer J 2025;57(4):385–94. https://doi.org/10.1038/s41428-024-00997-4.
[73] Zheng T, Li T, Shi J, Wu T, Zhuang Z, Xu J, Guo B. Molecular insight into the toughness of polyureas: a hybrid all atom/coarse grained molecular dynamics study. Macromolecules 2022;55(8):3020–9. https://doi.org/10.1021/acs.macromol.1c02453.
[74] Ma C, Ji T, Robertson CG, Rajeshbabu R, Zhu J, Dong Y. Molecular insight into the Mullins effect: irreversible disentanglement of polymer chains revealed by molecular dynamics simulations. Phys Chem Chem Phys 2017;19(29):19468–77. https://doi.org/10.1039/C7CP01142C.
[75] Feng S, Qi L, Wang L, Pan S, Ma M, Zhang X, Li G, Liu R. Atomic structure of shear bands in Cu64Zr36 metallic glasses studied by molecular dynamics simulations. Acta Mater 2015;95:236–43. https://doi.org/10.1016/j.actamat.2015.05.047.
[76] Qin Z, Yang Y, Mi HY, Tian Q, Jing X, Antwi-Afari MF, Dong B, Liu C, Shen C. Hierarchical H-bonding and metal coordination bonds enabled supramolecular dual networks for high-performance energy-dissipation. Chem Eng J 2024;498:155414. https://doi.org/10.1016/j.cej.2024.155414.
[77] Guo K, Qu J, Chen R, Liu Q, Yu J, Liu J, Zhu J, Sun G, Liu P, Wang J. A hierarchical energy dissipation strategy for impact-resistant polyurea driven by multiple hydrogen bonds. Constr Build Mater 2025;487:142056. https://doi.org/10.1016/j.conbuildmat.2025.142056.
[78] Zahra M, Javed M, Iqbal S, Mahmood S, Sarfraz S, Yasin M, Alshalwi M. High-strength montmorillonite polyurethane nanocomposites with exfoliated montmorillonite: preparation and characterization. Inorg Chem Commun 2024;162:112262. https://doi.org/10.1016/j.inoche.2024.112262.
[79] Qian M, Wang X, Yao L, Zhu Y. Enhanced mechanical and dielectric properties of polyurethane elastomers containing modified SiO2. ACS Omega 2024;9(47):47315–23. https://doi.org/10.1021/acsomega.4c08565.
[80] Tang J, Xu J, Fan H, Li R, Shi M, Liu Y, Wang B, Liu P. Preparation of hydroxy-terminated polydimethylsiloxane and nano-SiO2 hydrophobic polyurethane coated urea and investigation of controlled nitrogen release. Chem Eng Sci 2025;302:120728. https://doi.org/10.1016/j.ces.2024.120728.